

Group Assignment: Real-World Financial Time Series Analytics

Group size: 4 students

Objective

The objective of this assignment is to investigate a real-world financial forecasting problem using time-series analysis and machine-learning techniques. Each group is required to identify and formulate its own financial problem, obtain an appropriate real-world dataset, develop forecasting models, and critically evaluate the performance of traditional time-series and machine-learning approaches.

Task

Each group must:

1. Identify a real-world financial problem that can be investigated using time-series data.
2. Formulate a clear and specific research question.
3. Obtain a suitable real-world financial dataset from a reliable source.
4. Perform exploratory and statistical analysis of the selected time series.
5. Develop at least **two traditional time-series models** appropriate to the selected problem.
6. Develop at least **two machine-learning models** appropriate to the selected problem.
7. Include at least one simple baseline forecasting method for comparison.
8. Evaluate all models using an appropriate time-based train/validation/test strategy.
9. Compare the forecasting performance of the models using suitable evaluation measures.
10. Investigate at least one additional aspect of the problem, such as market shocks, feature importance, different forecasting horizons, volatility regimes, model stability, or model interpretability.
11. Interpret the results from a financial or business perspective.
12. Discuss the limitations of the analysis and identify possible improvements.

Important Requirements

- The selected problem must have a clear real-world financial application.
- The dataset must be based on actual financial observations.
- Random train-test splitting should not be used for time-series forecasting.
- The choice of each model must be justified based on the characteristics of the data and the research problem.
- Students should not simply compare error values. They must explain why the models produced different results.
- The analysis must include appropriate visualizations, statistical tests, model diagnostics, and forecast evaluation.
- All code and data-processing steps must be documented sufficiently to reproduce the results.

Possible Areas

Students may consider areas such as:

- Stock price or return forecasting
- Stock-market index forecasting
- Exchange-rate forecasting
- Cryptocurrency forecasting
- Gold and commodity-price forecasting
- Interest-rate or bond-yield forecasting
- Trading-volume forecasting
- Financial volatility forecasting
- Portfolio risk forecasting
- Value-at-Risk forecasting
- Multivariate financial forecasting
- Other relevant financial time-series problems

These are examples only. Groups are encouraged to identify their own interesting financial problem.

Required Modelling Perspective

The project must include both:

Traditional Time-Series Approach

Examples include AR, MA, ARMA, ARIMA, SARIMA, GARCH and related models.

Machine-Learning Approach

Examples include Linear Regression, Random Forest, XGBoost, Support Vector Regression, Neural Networks, LSTM and related models.

Students should select models that are appropriate for their specific problem rather than attempting to use every available model.

Deliverables

1. Report:

Core Question

The project should ultimately answer:

How effectively can the selected financial problem be forecast using traditional time-series methods compared with machine-learning methods, and what can be learned from the differences in their performance?